Cambridge (CIE) A Level Computer Science

Your notes

Program Development Life cycle

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Lifecycle stages

What is the program development life cyle?

- The **program development life cycle (PDLC)** is a structured process used to **design, build, and test software solutions**
- It ensures that programs are developed **systematically**, **efficiently**, and to a **high standard**
- Each stage of the cycle has a **specific purpose**, and together they help developers **solve real-world problems** with reliable software

Analysis

- The **analysis stage** is all about **understanding the problem** the program is being created to solve
- Developers use **abstraction** to ignore unnecessary detail and focus only on what matters:
  - What the program **must do** (core functionality)
  - What **limitations** it must work within (constraints)
- The London Underground map is a great example of abstraction, you don't need the real geography, just a clear route from stop A to stop B

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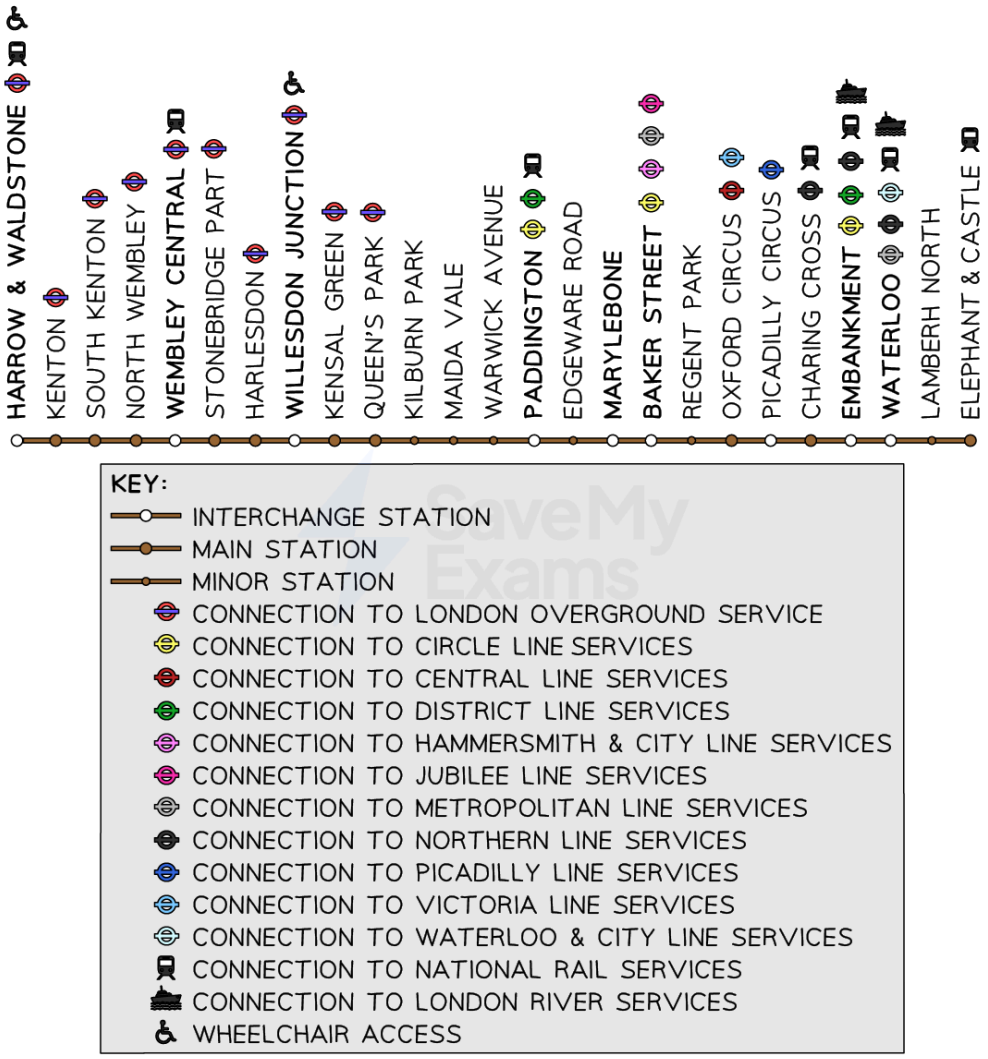

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- At this stage, a **requirements document** is often created. It:
  - **Breaks** the problem into manageable parts
  - **Labels** each requirement
  - **Describes** what success looks like for each feature

Design

- The **design stage** involves planning how the program will work
- Developers create a **blueprint** for the solution using tools such as:
  - **Structure diagrams** – break down the program into smaller components
  - **Flowcharts** – show the logical flow of processes
  - **Pseudocode** – outlines how the logic will be written in code

Coding

- In the **coding stage**, developers begin writing the program using a **suitable programming language**
- Code is written in **modules** that work together to solve the full problem
- **Iterative testing** is used – each module is tested and debugged individually as it's created
- Modules are **retested** whenever changes are made, to ensure no new errors are introduced

Testing

- Once the full program is written, it is thoroughly tested using a variety of **test data** to ensure:
  - It meets all the original requirements
  - It handles valid and invalid input correctly
  - It performs reliably under different conditions
- **Example test data**: Alphanumeric sequences used to check password input validation

Maintenance

- After the program has been delivered to the client or end users, it enters the **maintenance stage**
- This involves:
  - **Fixing bugs** that weren't discovered during testing
  - **Updating features** to meet new requirements
  - **Improving performance** or adapting to new systems
- There are three main types of maintenance:
  - **Corrective** – fixing errors
  - **Adaptive** – updating the software to run on new hardware or platforms
  - **Perfective** – improving features based on user feedback
- Software maintenance ensures the program **remains useful and reliable over time**

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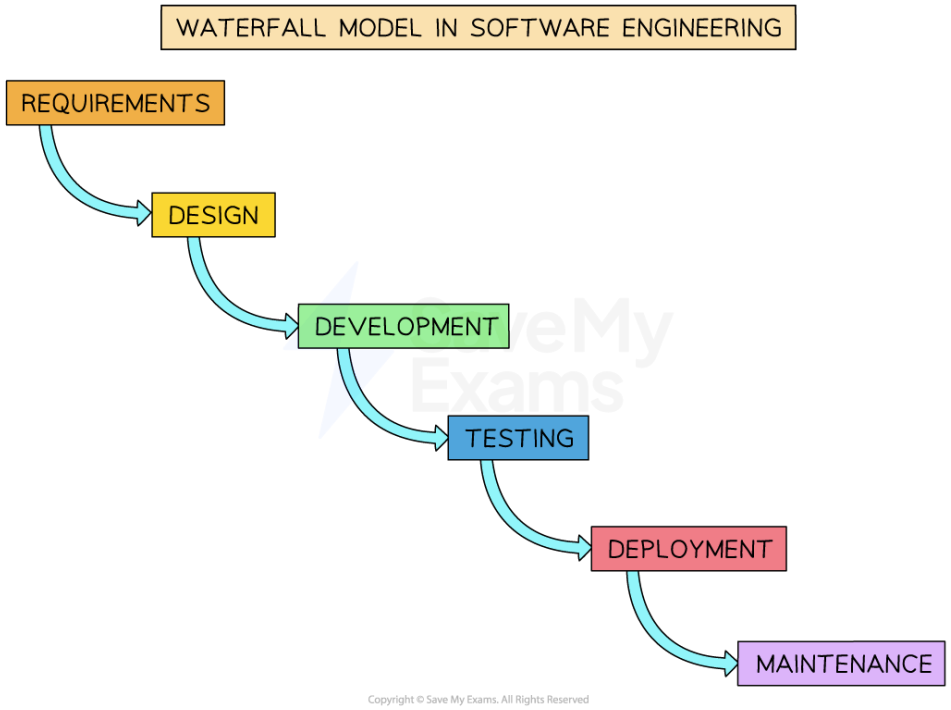
Waterfall

What is the waterfall model?

- The Waterfall Model is a **sequential software development process** divided into **distinct phases**
- Each phase **must be completed** before the next one begins

Steps

1. **Requirement Gathering and Analysis:** All possible system requirements to be developed are captured and documented clearly
2. **System Design:** The requirements are translated into a design. Architects and designers define the overall architecture and identify the main components
3. **Implementation:** The actual code is written in this phase based on the design documents, turning the system design into a functional program
4. **Integration and Testing:** All the components and modules are integrated and tested to ensure that the entire system works as expected
5. **Deployment:** The product is released to the market or handed over to the client. It may involve installation, customization, and training
6. **Maintenance:** Post-release, the system needs regular maintenance to fix bugs, improve performance, or add new features



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Benefits and drawbacks

Benefits	Drawbacks
Simple and linear – easy to understand and follow	**Inflexible** – difficult to make changes once development begins
Clear stages and milestones – easy to track progress	**Expensive to fix late problems** – issues found late are harder to resolve
Ideal for well-defined projects – works best when requirements are fixed	**Long development cycle** – each stage must be completed before moving on

Suitability

- The Waterfall Model is most suitable for projects where **requirements are well understood** and **unlikely to change**
- It works well when **high quality and compliance are essential**, and there is a clear understanding of the project's goals and constraints

Iterative (Agile)

What is the iterative model?

- The iterative model s a **type of Agile software development methodology** that promotes **adaptability** and high customer involvement

Steps

1. **Identify user stories and requirements**
    1. Work closely with stakeholders to gather functional and non-functional requirements
    2. Requirements are often written as **user stories** (e.g. *As a user, I want to...*)
2. **Plan the sprint (Sprint Planning)**
    1. Break down requirements into **tasks**
    2. Choose a set of tasks (features) for the **current sprint** (a short time-boxed development period, usually 1–4 weeks)
    3. Define the **sprint goal**
3. **Design the solution**
    1. Decide how the selected features will be built
    2. Focus is on **simple and adaptable design**, not heavy upfront documentation

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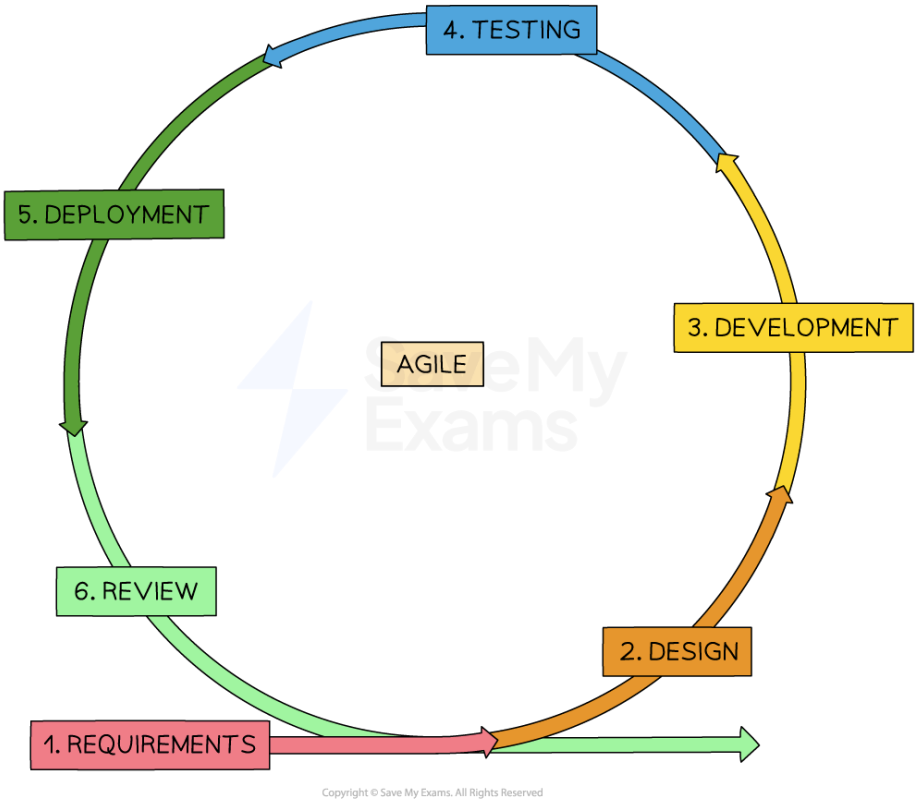
4. **Develop the features**
    1. Write code for the selected tasks in the sprint
    2. Developers often work in pairs or small teams (e.g. **pair programming**)
5. **Test continuously**
    1. Perform **unit testing**, **integration testing**, and **acceptance testing** during the sprint
    2. Testing is **ongoing**, not saved for the end
6. **Review progress (Sprint Review)**
    1. Demo the working software to stakeholders
    2. Collect feedback and identify changes or improvements
7. **Reflect on process (Sprint Retrospective)**
    1. The team reflects on what went well and what needs improving in the next sprint
    2. Focus is on **team performance and process optimisation**
8. **Release** (may happen after every sprint or set of sprints)
    1. Deploy working software to users or staging environment
9. **Repeat**
    1. Move to the next sprint with updated priorities and feedback



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Benefits and drawbacks

Benefits	Drawbacks
Highly adaptable – responds quickly to changing requirements	**Requires experienced team members** – may be hard to manage without expertise
Frequent communication – promotes constant collaboration	**Risk of burnout** – intense collaboration can tire the team
Focus on quality – encourages good design and continuous testing	**May lack documentation** – flexibility can reduce written records
Customer collaboration – ensures the product meets real needs	**Scope creep** – changing goals may lead to uncontrolled growth

Suitability

- The iterative model is most suitable for **small to medium-sized projects** where **requirements can change** and customer involvement is high

Rapid Application Development (RAD)

What is RAD?

- Rapid Application Development (**RAD**) is a software development methodology that **emphasises fast and iterative development**

Steps

1. **Requirement planning:** Gather general system requirements, define constraints and assumptions
2. **User design and prototyping:** Collaborate with users to develop prototypes, ensuring alignment with user needs
3. **Construction or iterative development:** Build the system incrementally, with continuous user feedback and adaptation
4. **Cutover or deployment:** Transition the product into the live environment, including user training, support, and documentation
5. **Maintenance and updates:** Continue to adapt and improve the system based on user feedback and needs

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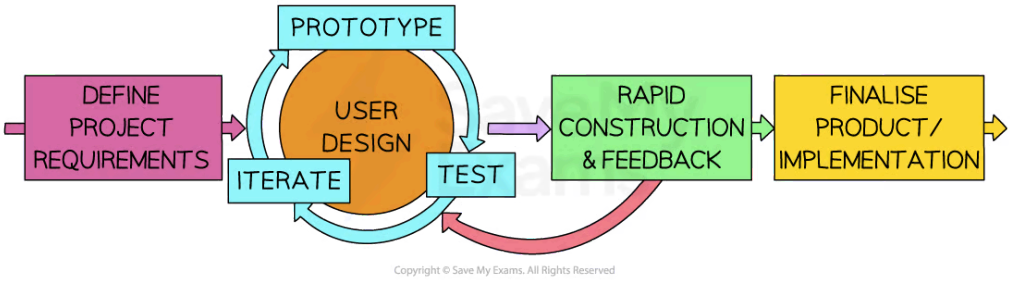

Rapid Application Development (RAD) Model of Software Development

Benefits and drawbacks

Benefits	Drawbacks
Speed – fast development and delivery at relatively low cost	**Requires strong team collaboration** – skilled and cohesive teams are essential
User involvement – client feedback shapes the system throughout	**Potential quality issues** – speed may reduce testing and documentation
Flexibility – adapts quickly to changing requirements	**Not ideal for small projects** – may be too complex for simple systems
Incremental development – builds in small, testable parts	**Scope creep risk** – flexibility can lead to ever-expanding requirements

Suitability

- Rapid Application Development is most suitable for projects where **rapid delivery is required** and where requirements can be developed and **refined on the go**

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